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# The v7 Pipeline: Manuscript to Illustrated Book

Stage-by-stage flow - reference sheets, fact contracts, measured gates, diagnostic repair

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# The v7 Pipeline — How a Manuscript Becomes an Illustrated Book

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One sentence: **manuscript -> plan -> QA'd reference sheets -> per-page fact contracts -> multi-reference generation (sheets + style plate as images) -> scored yes/no verification -> targeted repair or flag -> typeset -> PDF.**

The philosophy shift from the previous pipeline: character identity and art style used to be requested in prompts and repaired afterwards; now they are supplied as reference images and measured before a page is accepted.

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## Stage 1 — Parse (docx -> plan)

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We read the client's manuscript docx and have an LLM extract the whole book plan as structured data: the style description, every character with a locked spec (species, colours, signature items like "green cap, red baseball bandana — always"), look-alike groups (Bilbo + Obi), recurring settings, and one scene per page (setting, action, mood, and the printed page text). Saved as `.toon` files under the version dir (e.g. `v6_b/data/`).

## Stage 2 — Reference sheets (the "Book Bible")

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For each main character we generate an official character sheet (portrait + full body on a white background); for the look-alike duo a combined sheet showing both side by side, so their contrast — green vs blue cap — is defined in one image; and a plate per recurring setting. When the client provides real art (their V6 PDF pages), we condition on it for likeness. Every sheet is QA'd by a vision judge and re-rendered until it matches the spec.

These sheets become the single source of truth: everything downstream copies them, never a fresh interpretation of the text.

## Stage 3 — Layout

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An LLM decides each page's composition: where each character sits (coordinate boxes) and which side of the frame stays empty for the caption text.

## Stage 4 — Page contracts (new in v7)

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From the printed text of each page we extract verifiable facts: "exactly 2 adults", "exactly 2 pup cups", "the dogs must look sleepy". Each contract is used twice — written into the generation prompt, and checked against the finished art. Generation and verification can never disagree about what was required.

## Stage 5 — Generate (the big v7 change)

For every page — story pages AND cover / title / dedication / about-author — we call Gemini's image model with the scene prompt plus a stack of reference images, each with a named job:

- "Image 1 = Bilbo & Obi's official designs — copy each EXACTLY."
- "Image 2 = the book's ART STYLE — match it; don't copy its scene."
- "Image 3 = the recurring location — same place, same colours."

Identity and style come from images, not from words. Because the style plate rides on every page, matter pages can no longer drift photorealistic.

(The previous way was FLUX Kontext + trained LoRA weights. A measured bake-off on the six hardest pages showed raw Gemini multi-ref beats even the corrected Kontext output — so LoRA training was removed from the flow: cheaper, faster, and one fewer copy-of-a-copy in the identity chain.)

## Stage 6 — Measured gate (new in v7)

Instead of a judge saying "looks fine", we ask a vision model batches of yes/no questions and score each category 0–100:

- **Identity** — same muzzle/head shape, same signature items, same apparent age as the reference sheet?
- **Facts** — exactly two pup cups? exactly two adults? do the dogs look sleepy? (straight from the page contract)
- **Anatomy** — every limb attached and counted, no human legs on animals, no duplicated characters, no text inside the art?
- **Style** — same rendering family as the style plate, not photorealistic?

Binary questions anchored to spec facts leave the judge nothing to invent — this replaced a prose judge that once hallucinated a collar that existed in no spec and no sheet.

## Stage 7 — Diagnostic repair

Each failure type gets its own remedy (never "same prompt, roll again"):

| Failure          | Remedy                                           |
|------------------|--------------------------------------------------|
| anatomy or style | REGENERATE the page — structure can't be patched |

| Failure               | Remedy                                                                            |
|-----------------------|-----------------------------------------------------------------------------------|
| identity or facts     | surgical i2i edit quoting the exact failed questions ("remove the third pup cup") |
| caption band occupied | repaint just that band into calm sky/ground                                       |

Two repair rounds maximum. A page that still fails keeps its best-scoring attempt and is **flagged** in `gate_report.json` for human review — surfaced honestly instead of shipped silently.

## Stage 8 — Compose + book

Typesetting: find the calmest strip on the page, place the caption there with adaptive ink (dark or light) and a soft scrim when the background needs it — then bind all pages into the print-ready PDF.

## Old pipeline vs v7, at a glance

|                     | v6 (kontext + LoRA)                   | v7                                   |
|---------------------|---------------------------------------|--------------------------------------|
| identity carried by | trained LoRA weights + stitched sheet | role-assigned reference images       |
| style carried by    | prompt text (nothing on matter pages) | style plate image on EVERY page      |
| story facts         | not extracted, not checked            | contract: in the prompt AND verified |
| quality check       | prose vision judge                    | scored yes/no probes per category    |
| on failure          | same-prompt retry lottery             | remedy matched to the failure type   |

| v6 (kontext + LoRA) |                        | v7                       |
|---------------------|------------------------|--------------------------|
| unfixable page      | shipped anyway         | flagged for human review |
| LoRA training       | ~\$6 + 20 min per book | none                     |